

# 02\_FINAL\_REQUIREMENTS\_SPECIFICATION.md

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## SIH26034 - Legal Metrology Automated Compliance System

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### Master Requirements Specification (MRS) - Functional & Non-Functional Requirements with Verification Criteria

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#### 1. Statutory Context & Legal Authority

This Master Requirements Specification defines the precise technical and legal requirements for **SIH26034: Software System to check compliance of Packaged Commodities under Legal Metrology (Packaged Commodities) Rules, 2011** by scanning products, images and labels.

The system derives its statutory authority and testing thresholds from:

**1. The Legal Metrology Act, 2009 (Act No. 1 of 2010):**

- Section 18: Mandatory adherence to standard quantities and packaging declarations.
- Section 36(1): Penal provisions and compounding fines for manufacturing, packing, selling, or distributing non-compliant commodities.

**2. The Legal Metrology (Packaged Commodities) Rules, 2011:**

- Rule 6(1): Mandatory on-pack statutory declarations.
- Rule 6(10) & Rule 6(10A): E-commerce platform obligations and Country of Origin search/filtering mandates (as amended by G.S.R. 128(E) dated 13.02.2026, effective 1 July 2026).
- Rule 7 & Table-I: Minimum font size and numeral height standards as a function of Principal Display Panel (PDP) area (amended by G.S.R. 629(E)

dated 23.06.2017).

- Rule 6(1)(k): Unit Sale Price (USP) computation and display requirements (amended by G.S.R. 779(E) dated 02.11.2021).

3. The Bharatiya Sakshya Adhiniyam, 2023 (BSA 2023):

- Section 63: Admissibility of electronic records in legal proceedings (replacing repealed Section 65B of the Indian Evidence Act, 1872).

## 2. Functional Requirements (FR-01 to FR-20)

| +-----+<br>-----+<br>  REQ ID   REQUIREMENT TITLE   STATUTORY BASE   ACCEPTANCE CRITERIA<br> <br>+-----+<br>-----+ |                                                    |                    |                     |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------|---------------------|
| FR-01                                                                                                              | Forensic Image Ingestion & SHA-256 computed before | BSA 2023 Sec 63    | Image ingested,     |
|                                                                                                                    | SHA-256 Hashing                                    |                    | any transformation, |
|                                                                                                                    | logged to ledger.                                  |                    |                     |
| +-----+<br>-----+                                                                                                  |                                                    |                    |                     |
| FR-02                                                                                                              | Optical Quality Gating                             | Technical Standard | Rejects blur (Var < |
|                                                                                                                    | 150), glare (> 3%),                                |                    |                     |
|                                                                                                                    |                                                    |                    | tilt (> 15°); emits |
|                                                                                                                    | real-time HUD advice.                              |                    |                     |
| +-----+<br>-----+                                                                                                  |                                                    |                    |                     |
| FR-03                                                                                                              | Fiducial Calibration &                             | Technical Standard | Detects ArUco 4x4 / |
|                                                                                                                    | coin; computes exact                               |                    |                     |
|                                                                                                                    | Perspective Rectification                          |                    | px_to_mm scale;     |
|                                                                                                                    | warps perspective.                                 |                    |                     |
| +-----+<br>-----+                                                                                                  |                                                    |                    |                     |
| FR-04                                                                                                              | Principal Display Panel                            | LM (PC) Rule 2(h)  | Calculates PDP area |
|                                                                                                                    | based on package                                   |                    |                     |
|                                                                                                                    | (PDP) Area Calculation                             | & Rule 7           | geometry (40% rule  |
|                                                                                                                    | for rect/cyl).                                     |                    |                     |
| +-----+<br>-----+                                                                                                  |                                                    |                    |                     |
| FR-05                                                                                                              | Multi-Oriented Text Detection                      | Technical Standard | DBNet++ extracts    |
|                                                                                                                    | text polygon bounding                              |                    |                     |
|                                                                                                                    |                                                    |                    | boxes with IoU >=   |
|                                                                                                                    | 0.50.                                              |                    |                     |
| +-----+<br>-----+                                                                                                  |                                                    |                    |                     |
| FR-06                                                                                                              | Optical Character Recognition                      | Technical Standard | PaddleOCR PP-OCrv4  |
|                                                                                                                    | transcribes text;                                  |                    |                     |
|                                                                                                                    | (OCR)                                              |                    | Character Error     |
|                                                                                                                    | Rate (CER) <= 2.5%.                                |                    |                     |
| +-----+<br>-----+                                                                                                  |                                                    |                    |                     |

|                   |                                                   |                      |                                            |
|-------------------|---------------------------------------------------|----------------------|--------------------------------------------|
| FR-07             | Manufacturer/Packer/Importer                      | LM (PC) Rule 6(1)(a) | Validates presence of registered corporate |
|                   | Address Extraction                                |                      | identity, city, state, and pin code.       |
| +-----+<br>-----+ |                                                   |                      |                                            |
| FR-08             | Net Quantity & Table-I Font                       | LM (PC) Rule 6(1)(h) | Measures numeral x-height; cross-checks    |
|                   | Height Verification                               | & Table-I (GSR 629E) | against PDP area threshold (incl. 6.0mm).  |
| +-----+<br>-----+ |                                                   |                      |                                            |
| FR-09             | Maximum Retail Price (MRP) &                      | LM (PC) Rule 6(1)(e) | Validates currency, numeral value, and     |
|                   | Tax Inclusivity Check                             |                      | mandatory "(incl. of all taxes)" clause.   |
| +-----+<br>-----+ |                                                   |                      |                                            |
| FR-10             | Unit Sale Price (USP)                             | LM (PC) Rule 6(1)(k) | Verifies presence, correct unit denom,     |
|                   | Mathematical Cross-Check                          | (GSR 779E)           | and math:  USP_calc - USP_decl  <= 0.01.   |
| +-----+<br>-----+ |                                                   |                      |                                            |
| FR-11             | Date of Manufacture / Expiry                      | LM (PC) Rule 6(1)(d) | Extracts month & year of manufacture /     |
|                   | Verification                                      |                      | packaging; checks format compliance.       |
| +-----+<br>-----+ |                                                   |                      |                                            |
| FR-12             | Consumer Care Contact                             | LM (PC) Rule 6(1)(n) | Validates presence of person/office name,  |
|                   | Verification                                      |                      | telephone number, and valid email format.  |
| +-----+<br>-----+ |                                                   |                      |                                            |
| FR-13             | Country of Origin Extraction                      | LM (PC) Rule 6(1)(p) | Detects origin declaration; flags missing  |
|                   |                                                   |                      | origin on imported packaged goods.         |
| +-----+<br>-----+ |                                                   |                      |                                            |
| FR-14             | E-Commerce Mandatory checks digital presence      | LM (PC) Rule 6(10)   | Ingests DOM/URL;                           |
|                   | Attribute Extraction                              |                      | of all Rule 6 statutory declarations.      |
| +-----+<br>-----+ |                                                   |                      |                                            |
| FR-15             | E-Commerce 2026 Country of origin is exposed as a | LM (PC) Rule 6(10A)  | Checks whether                             |
|                   | Origin Filter Audit                               | (GSR 128E)           | structured, searchable, sortable filter.   |
| +-----+<br>-----+ |                                                   |                      |                                            |
| FR-16             | E-Commerce vs Packaging between webpage text      | Consumer Protection  | Flags discrepancies                        |
|                   | Discrepancy Cross-Check                           | Act 2019 / LM Act    | and physical packaging gallery image.      |
| +-----+           |                                                   |                      |                                            |

|       |                             |                       |                                            |
|-------|-----------------------------|-----------------------|--------------------------------------------|
| FR-17 | Interactive Adjudication    | Principles of         | Dual-pane canvas; synchronized pan/zoom;   |
|       | Canvas & Pixel Loupe        | Natural Justice       | interactive pixel loupe with mm display.   |
| FR-18 | Mandatory Officer Override  | Evidence Procedure    | Any modification of AI findings requires   |
|       | Justification Logging       |                       | officer PIN and mandatory text remarks.    |
| FR-19 | Statutory Legal Notice      | LM Act 2009 Sec 36(1) | Generates archival PDF/A Form-1/2 notice   |
|       | Compilation                 | Form-1 / Form-2       | with embedded photographic evidence crops. |
| FR-20 | Section 63 BSA 2023 Digital | BSA 2023 Sec 63       | Emits electronic certificate with device   |
|       | Evidence Certification      |                       | metadata, image hashes, officer signature. |

- ### 3. Non-Functional Requirements (NFR-01 to NFR-10)

  - NFR-01: End-to-End Pipeline Latency:**
    - Total execution time from raw image receipt to completed rule evaluation must be  $\leq 1200$  ms on an 8-core CPU without GPU acceleration.
  - NFR-02: Physical Font Measurement Precision:**
    - Mean Absolute Error (MAE) of measured font x-height must be  $\leq 0.30$  mm when benchmarked against certified digital vernier calipers on planar surfaces.
  - NFR-03: Text Recognition Accuracy:**
    - Character Error Rate (CER) on statutory declaration fields must not exceed 2.5% under standard lighting conditions ( $\geq 300$  lux).
  - NFR-04: Intellectual Property & Licensing Safety:**
    - 100% of third-party libraries, models, and dependencies must carry permissive licenses (Apache-2.0, MIT, BSD). Zero AGPL-3.0 or GPL-3.0 copyleft dependencies permitted.
  - NFR-05: Evidentiary Admissibility:**

- All exported digital records, certificates, and evidence bundles must conform strictly to the statutory evidentiary requirements of Section 63 of the Bharatiya Sakshya Adhiniyam, 2023.

6. **NFR-06: Zero Autonomous False Accusations:**

- Under no circumstances shall an automated penalty or legal notice be issued without explicit, authenticated human officer adjudication. False Accusation Rate = 0.0%.

7. **NFR-07: Offline Field Resilience:**

- The field inspection and capture module must function reliably without active internet connectivity using local SQLite/SQLCipher caching.

8. **NFR-08: Cryptographic Tamper-Evidence:**

- The audit log must use SHA-256 cryptographic chaining (Merkle Tree architecture). Any alteration, insertion, or deletion of inspection rows must invalidate subsequent hash verification.

9. **NFR-09: Accessibility & Usability:**

- The user interface must comply with WCAG 2.1 AA standards, supporting keyboard navigation, high-contrast display modes, and touch target sizes  $\geq$  48 dp.

10. **NFR-10: Statutory Data Retention:**

- The data tier must enforce a minimum 7-year immutable retention policy for all inspection records, raw images, and issued legal notices.

# 4. Requirements Traceability Matrix (RTM Sample)

| +-----+<br>-----+<br>  REQ ID   ARCHITECTURE COMPONENT   DATABASE TABLE   TEST SUITE<br>        |  |  |  |
|-------------------------------------------------------------------------------------------------|--|--|--|
| +-----+<br>-----+<br>  FR-01   Ingestion / Hashing Service   evidence_images   TS-01 (Unit)<br> |  |  |  |
| FR-02   Optical Quality Gate   evidence_images   TS-02 (Unit)<br>                               |  |  |  |
| FR-03   Fiducial Calibration Engine   evidence_images   TS-03 (Unit)<br>                        |  |  |  |
| FR-04   PDP Geometry Calculator   inspections   TS-04 (Unit)<br>                                |  |  |  |
| FR-05   DBNet++ Text Detection   bounding_boxes   TS-05 (Model)<br>                             |  |  |  |
| FR-06   PaddleOCR PP-OCRv4 Engine   bounding_boxes   TS-06 (Model)                              |  |  |  |

|               |                                |                        |             |
|---------------|--------------------------------|------------------------|-------------|
|               |                                |                        |             |
| FR-08         | Physical Font Measurement      | compliance_evaluations | TS-07       |
| (Integration) |                                |                        |             |
| FR-10         | USP Deterministic Calculator   | compliance_evaluations | TS-08       |
| (Integration) |                                |                        |             |
| FR-17         | Adjudication Canvas (Frontend) | inspections            | TS-15 (E2E) |
|               |                                |                        |             |
| FR-19         | ReportLab PDF Notice Generator | legal_notices          | TS-18       |
| (System)      |                                |                        |             |
| FR-20         | Section 63 BSA Cert Generator  | bsa_certificates       | TS-19       |
| (Compliance)  |                                |                        |             |
| +-----        |                                |                        |             |
| -----+        |                                |                        |             |